Orbital Dragons to Hunt Cryptids: Utilizing Non-Euclidean Geometry to Solve Busy Beaver

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DeepMath 2026 — Extended Abstract

Abstract

Whether the sixth Busy Beaver value can be determined turns on a residual set of "cryptid" Turing machines whose halting problems have resisted every decider brought to bear on them. We reconceptualize the halting question for this class as an event-history problem: the machine's evolution is a long-duration process punctuated by recurrent events, and halting is a terminal event whose hazard concentrates in identifiable regions of a low-dimensional state space. Exploiting this structure, we develop a certified pipeline in which raw tape dynamics are aggregated into orbit events, embedded in non-Euclidean (logarithmic, simplex, hyperbolic) coordinate systems where the dynamics become affine, and then re-derived as machine-checked theorems in core Lean 4 under a minimal axiom set. Our central result reduces two previously open BB(6) machines, end to end, to conditional theorems of the form $\mathsf{SafeMetaOrbit} \to \neg\mathsf{RawHalts}$, in which the antecedent concerns only a deterministic affine map on five integers. We validate each layer by differential testing against 10^9 raw machine steps and show that the excluded parameter configurations are lethal: entering them provably produces the halting transition. The remaining conjecture is therefore not an artifact of the method but the halting problem itself, compressed to arithmetic. We close by situating the approach within the formal-mathematics workflow emerging around Lean and large language models.

1 Motivation and framing

The determination of $BB(5) = 47{,}176{,}870$, machine-checked in Rocq by the bbchallenge Collaboration [1, 2], transformed the Busy Beaver program from a computational pastime into a subfield of verified mathematics. The frontier now rests at $n = 6$, where the obstruction is qualitative rather than quantitative: a small residual set of machines—the cryptids, of which Antihydra is the best-known—encode Collatz-like arithmetic, so that deciding them appears to require number theory beyond current reach [3]. Tao's result that almost all Collatz orbits attain almost bounded values [4] marks the present boundary of what is provable for such maps, and it is a measure-theoretic statement; the Busy Beaver question demands pointwise control of a single orbit.

Our point of departure is methodological. Treating the machine as a dynamical system in the tradition of Moore and Kůrka [5, 6], we observe that for the compact-growth cryptids the hazard of halting is not diffuse in time: it concentrates at *incision events*, rare configurations in which a distinguished probe state reads the one symbol that could initiate the halting transition. The natural framework is that of event-history analysis: long censored spells of routine motion, recurrent events with measurable covariates, occasional regime transitions (renewals), and a terminal event whose risk set we can characterize exactly. The methodological task is then to certify (i) that the risk set is where we say it is, and (ii) what the process does between visits to it.

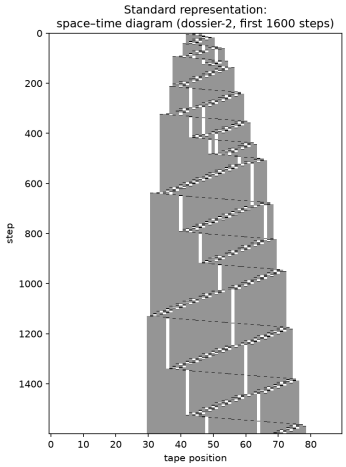


Figure 1: The standard representation: a space–time diagram of the first 1,600 steps of dossier-2. Texture is visible; structure is not.

2 Data reduction: from tapes to orbits to charts

Figure 1 displays the standard representations of one of our two target machines: the space–time diagram and the tape-extent growth curves. Both are faithful and both are uninformative; the self-similar structure of the process is invisible at this level of aggregation.

We therefore aggregate. Record-level analysis of the tape yields, for each machine, a shape vector: the field boundaries and the positions of two mobile "holes," summarized as counters (a, b, c). Embedding the resulting orbit in three non-Euclidean charts—the componentwise logarithmic map of Maslov–Litvinov dequantization [7, 8], the projective simplex normalization of the Perron–Frobenius tradition, and a hyperbolic shape-versus-scale chart in the spirit of Poincaré embeddings [9]—renders the dynamics affine (Figure 2). Generic evolution appears as line segments; renewals appear as corners and jumps. It is this piecewise affinity, discovered geometrically, that the certification layer then proves.

3 Certification: from charts back to theorems

The geometric layer guides but does not warrant. Every claim used in the final theorems is rederived in core Lean 4, with axioms restricted to $\{\text{propext}, \text{Quot.sound}\}$, over the raw transition-table semantics validated against the Rocq lineage by parser identities, differential execution traces, and a mechanized bisimulation with a zipper-tape model.

The certified account of the two machines' dynamics is an *interaction calculus*: a ten-step caterpillar gait crawls the field; static holes are crossed (+4) or terminated (+6) according to the parity of arrival; adjacent holes fuse (+9); the field edge executes a parity-dependent extension. Seven episode theorems per machine—each a complete trajectory between successive spawn entries, with exact step counts and probe safety—are compiled from this calculus by simulation-driven generators and glued by two inductions (an n-fold cycle lemma and a mixed sweep lemma whose parity constraint explains every shift constant). A five-integer abstract orbit metaStep dispatches the episodes; a transport theorem, a segment-induction keystone, and fully concrete startup certificates (235 and 2,044 steps, including one degenerate episode per machine) complete the chain.

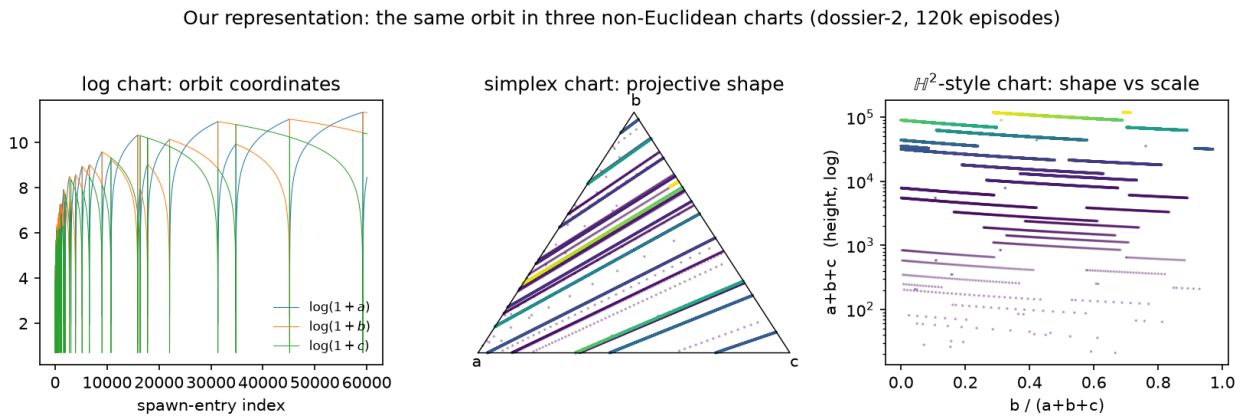


Figure 2: Our representation: 60,049 consecutive orbit events of dossier-2 in the three charts. Left: logarithmic chart; generic loops are affine segments. Center: simplex chart; scale is quotiented out and only shape dynamics remain. Right: hyperbolic-style chart; the eternal orbit is an ascending walk, one zig per renewal generation.

Theorem 1. *For each of the two machines, at axioms* {propext, Quot.sound}*:* SafeMetaOrbit $\rightarrow$ $\neg$ RawHalts(blankStart)*.*

4 Validation and the status of the conjecture

Three validation regimes discipline the pipeline. First, the empirical calculus is fail-closed: it must classify every episode exactly, and it does so for 10^9 raw steps per machine with no unexplained events. Second, the Lean metaStep is mirrored in Python and checked snapshot-for-snapshot against raw emulation, a test now permanent in the repository. Third, the abstract orbits are run to 10^8 episodes (a raw-step horizon near 10^{15}) without leaving the guarded region.

The substantive status of SafeMetaOrbit deserves emphasis (Figure 3). The renewal map is a subtractive Euclid–Collatz hybrid: generic loops conserve $\gamma = c - b$ while a advances; deep renewals exchange the roles of a and b; fusions rotate $(a, 2, c) \mapsto (a{+}9, c{-}6, 1)$. We verified by direct construction that the excluded configurations halt within roughly fifty steps. The conjecture is therefore sharp: no weaker invariant suffices, and no stronger one is available without resolving the arithmetic. What has changed is the type of the open problem—from a question about a Turing machine to a question about a five-integer recurrence.

5 Relation to the emerging formal workflow

The development is an instance of the human–AI–kernel workflow that Tao and collaborators have articulated for the Equational Theories Project and the Palomar registry [10, 11]: generators (here, an AI assistant compiling proofs from validated empirical calculi) are labor; the Lean kernel is the sole trust surface. Our experience adds two proof-engineering observations of possible independent use: `omega` imports classical choice through negated-conjunction hypotheses, and rewriting on `stepN` goals triggers whole-run definitional evaluation—both eliminable by constructive De Morgan lemmas and term-mode composition, respectively.

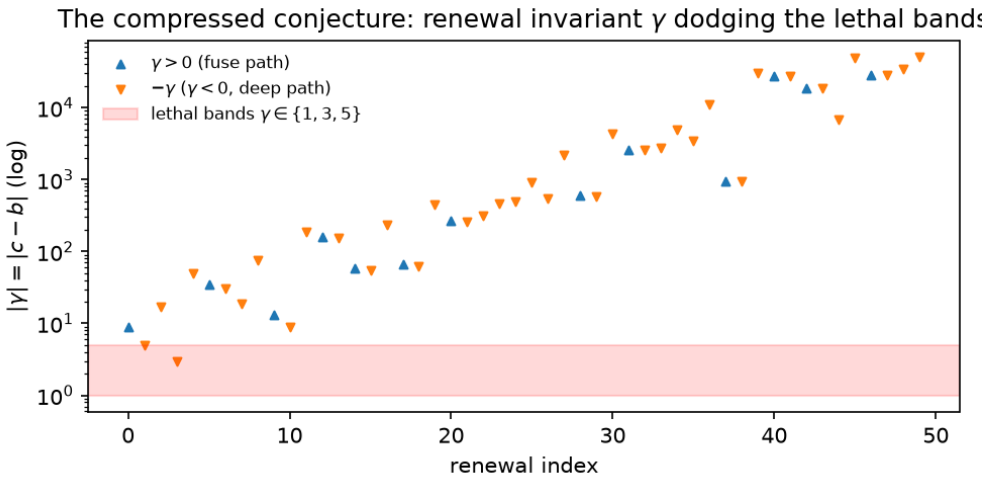


Figure 3: The compressed conjecture. The renewal invariant $\gamma = c - b$ across the first fifty renewals; the shaded band $\gamma \in \{1, 3, 5\}$ is provably lethal. SafeMetaOrbit asserts that this process never enters the band. This figure *is* the halting problem of dossier-2.

Honest scope. No unconditional non-halting claim is made for any open machine. The charts are instruments of discovery and exposition; no theorem depends on them. Empirical horizons are bounded by computation performed, not by what is true.

Acknowledgments. The Lean developments, empirical calculi, and proof compilers were built in collaboration with an AI assistant (Claude, Anthropic). All emitted proofs are checked by the Lean 4 kernel at the stated axiom bar.

References

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